

Technical Document #720: Retrieval Augmented Generation - Comprehensive Overview

Technical Document #720: Retrieval Augmented Generation - Comprehensive Overview

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Query decomposition breaks complex queries into simpler sub-queries.
- Hallucination detection identifies when LLMs generate factually incorrect information.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.